YEAR 3 MATHS

# Choosing Metric Units and Estimating

AC9M3M01 | Australian Curriculum v9.0

**Name:** ______________________ **Date:** ______________________

## Learning Intention

I can choose suitable metric units and use familiar benchmarks to estimate the length, mass or capacity of everyday items.

## Strategy Box: What are you measuring?

Length: millimetres (mm), centimetres (cm), metres (m), kilometres (km)

Mass: grams (g), kilograms (kg)     Capacity: millilitres (mL), litres (L)

**Choose a unit that gives a sensible number - not one that is extremely large or tiny.**

## Worked Example - Use a Benchmark

A classroom door is about 2 metres tall. Estimate: 2 m, not 2 cm or 2 km.

A full water bottle holds about 600 mL. This familiar item can be a benchmark.

## A. Warm Up - Choose the Best Unit

1. Length of a pencil: mm / cm / km

2. Distance between two towns: cm / m / km

3. Mass of an apple: g / kg

4. Capacity of a bucket: mL / L

5. Thickness of a coin: mm / m

6. Mass of a school bag: g / kg

## Picture Practice - Match the Tool

P1. Draw lines to match each measurement to a suitable tool.

| | |
|---|---|
| **Length of a desk** | measuring jug |
| **Mass of flour** | metre ruler |
| **Capacity of juice** | kitchen scales |

**Quick check: Unit + benchmark + reasonableness = a useful estimate.**

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## B. Build Your Skills - Make Sensible Estimates

7. A lunchbox is about ______ cm long.

8. A packet of rice has a mass of about ______ kg.

9. A bathtub holds about ______ L of water.

10. A paperclip is about ______ mm long.

## C. Select and Justify

11. Measure the classroom: Would you use cm, m or km? Explain.

Unit: ______ Because: ____________________________________________

12. Measure a teaspoon of medicine: Would you use mL or L? Explain.

Unit: ______ Because: ____________________________________________

13. Measure a watermelon: Would you use g or kg? Explain.

Unit: ______ Because: ____________________________________________

## D. Apply It - Estimate, Then Explain

14. Noah estimates that a swimming pool holds 500 mL. Is this reasonable?

Choose a better unit and explain your reasoning.

**15. Choose an object. Record what you would measure, the unit and your estimate.**

Object: __________ Measure: __________ Unit: ______ Estimate: __________

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# Choosing Metric Units and Estimating - Answer Key

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## Learning check

Accept reasonable estimates supported by suitable units and familiar benchmarks.
Exact values may vary.

## A. Warm Up and Picture Practice

1. cm 2. km 3. g 4. L 5. mm 6. kg

P1. Desk - metre ruler; flour - kitchen scales; juice - measuring jug.

## B. Sensible Estimates - Sample Answers

7. About 25 cm.

8. About 1 kg.

9. About 150 L.

10. About 30 mm.

## C-D. Select, Justify and Apply

11. m; a classroom is several metres long.

12. mL; medicine is a small liquid amount.

13. kg; a watermelon is relatively heavy.

14. Not reasonable. Use litres; a pool holds a very large liquid volume.

15. Answers vary; check the unit and estimate are sensible.

## Teacher feedback checklist

☐ Chooses a suitable unit ☐ Uses a familiar benchmark ☐ Estimates sensibly
☐ Explains the choice clearly ☐ Includes the unit with the number